

Exercise 1.1

```
[23]
✓ Os
import numpy as np
import pandas as pd
import os

if os.path.exists("week5_kigali_weather.csv"):
    df = pd.read_csv("week5_kigali_weather.csv")
else:
    df = pd.read_excel("week5_kigali_weather.xlsx")
    # Excel stores dates as serial numbers -> convert to YYYY-MM-DD text
    if np.issubdtype(df["date"].dtype, np.number):
        df["date"] = pd.to_datetime(df["date"], unit="D", origin="1899-12-30").dt.strftime("%Y-%m-%d")

dates = df["date"].astype(str).to_numpy()
temps = df["temp_c"].to_numpy(dtype=float)
rain = df["rainfall_mm"].to_numpy(dtype=float)
hum = df["humidity_pct"].to_numpy(dtype=float)
print(len(dates), "days loaded:", dates[0], "to", dates[-1])
```

60 days loaded: 2026-09-01 to 2026-10-30

Loading cell

```
]
Os
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print(len(dates), "days loaded:", dates[0], "to", dates[-1])
```

60 days loaded: 2026-09-01 to 2026-10-30

```
▶ fahrenheit = [t * 9/5 + 32 for t in temps_list]
print([round(f, 1) for f in fahrenheit])
```

Exercise 1.1

```
... [75.9, 78.6, 83.5, 66.7, 78.8, 71.8]
```

```
▶ hot = [t for t in temps_list if t >= 26]
cool_rounded = [round(t) for t in temps_list if t < 24]
print("hot:", hot)
print("cool_rounded:", cool_rounded)
```

Exercise 1.2

```
... hot: [28.6, 26.0]
cool_rounded: [19, 22]
```

```
[8]
Os
▶ raw_floats = [float(x) for x in raw]
print(raw_floats)
print("Average:", sum(raw_floats) / len(raw_floats))
```

... [24.4, 25.9, 28.6, 19.3]
Average: 24.55

Exercise 1.3

```
▶ results = [("Aline", 85), ("Eric", 78), ("Diane", 91), ("Samuel", 92), ("Grace", 76)]
ranked = sorted(results, key=lambda r: r[1], reverse=True)
print("Top 3:", ranked[:3])
```

... Top 3: [('Samuel', 92), ('Diane', 91), ('Aline', 85)]

Exercise 2.1

SCREENSHOTS OF LAB3

- Wettest day: 14 with 26.2 mm

```
... Python: 0.0644s | NumPy: 0.00415s | NumPy ~15x faster
```

```
... Mean: 24.00 °C
    Max: 28.6 °C
    Min: 19.3 °C
    Std: 2.61 °C
```

- Hottest day: 2026-09-17 at 28.6 °C

- ... (a) Number of hot days: 18
- ... (b) Average temperature on hot days: 27.11 °C

(a) Rainy days: 16
(b) Total rainfall: 331.2 mm
(c) Heaviest single-day rainfall: 26.2 mm

SCREENSHOTS OF LAB3

```
dry_mask = rain == 0
print(f"(a) Avg temp on rainy days: {temps[rainy_mask].mean():.2f} °C")
print(f"    Avg temp on dry days: {temps[dry_mask].mean():.2f} °C")
print("(b) Hot AND rainy days:", ((temps >= 26) & (rain > 0)).sum())
print("(c) Days with humidity >= 70:", (hum >= 70).sum())
```

... (a) Avg temp on rainy days: 24.14 °C
Avg temp on dry days: 23.95 °C
(b) Hot AND rainy days: 5
(c) Days with humidity >= 70: 16

Exercise4.3

```
anomal = ndarray: anomaly
print(ndarray with shape (60,))
print(f"(a) Days more than 3 °C above the mean:", (anomaly > 3).sum())
print("(b) Days more than 3 °C below the mean:", (anomaly < -3).sum())
```

... (a) Days more than 3 °C above the mean:
['2026-09-07' '2026-09-09' '2026-09-11' '2026-09-12' '2026-09-14'
 '2026-09-16' '2026-09-17' '2026-09-19' '2026-09-22']
(b) Days more than 3 °C below the mean: 8

Exercise5.1

```
first30 = temps[:30].mean()
last30 = temps[30:].mean()
print(f"First 30 days: {first30:.2f} °C")
print(f"Last 30 days: {last30:.2f} °C")
print(f"Difference: {last30 - first30:+.2f} °C")
# Comment: see the printed difference. Positive = Kigali got warmer, negative
```

... First 30 days: 26.21 °C
Last 30 days: 21.78 °C
Difference: -4.43 °C

Exercise5.2

```
top5 = sorted(zip(dates, rain), key=lambda p: p[1], reverse=True)[:5]
for d, r in top5:
    print(d, "-", r, "mm")
```

Bonus+3

... 2026-09-07 - 26.2 mm
2026-09-14 - 25.1 mm
2026-09-21 - 24.0 mm
2026-09-05 - 22.9 mm
2026-09-28 - 22.9 mm